

ZULHELMI SYAHTIAR

Fresh Graduate D3 Industrial Electronics

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SUMMARY

Fresh Graduate of Electrical Engineering, D3 Industrial Electronics program. Has a strong interest in the Internet of Things (IoT), microprocessors, programming, and control systems. Experienced in working on microcontroller-based projects and developing simple automation systems. Eager to keep learning, innovating, and applying the latest technology in the industrial world.

EXPERIENCE

PT PLN (Persero) ULTG Petukangan – South Tangerang

Jul 2025 – Nov 2025

Protection Maintenance Intern (HAR PRO)

- Assisted with routine maintenance and monitoring of substation protection systems.
- Tested protection relays (OCR, GFR, Differential, Distance Relay, OVR/UVR, Buchholz, Thermal Overload).
- Involved in system testing during transformer migration at a 150kV substation.
- Performed Battery Capacity Test (BCT) to maintain the reliability of the substation's backup power.
- Assisted with non-cascade testing on the incoming transformer panel.

Independent Clients – Jabodetabek

Mar 2023 – Present

Freelance Graphic Designer & Photographer

- Provided graduation and post-defense photography services, including style direction and photo editing.
- Designed print banners for clients' graduation celebration events.
- Created social media graphic designs for an organization's Instagram content.

EDUCATION

Politeknik Negeri Jakarta – Depok, West Java

2023 – 2026

D3 Industrial Electronics – GPA 3.55/4.00

- Final Project: "Dual DC Motor Control Trainer Kit – PID – Fuzzy Control Based on ESP32 with 3D Digital Twin Visualization".
- Teaching Assistant for a Community Service project installing a solar-powered Hydroponic Green House.
- Committee Member, PSYCHOROBOTIC Student Study Group, Informatics & Communication Department (2025).
- Committee member, Publication, Documentation & Creative Media division for E-TIME 2025, Olahraga Elektro 2024, FRACTION 2024 (Electrical Engineering Department Graduation), and the Industrial Electronics Study Excursion.
- Committee member, Media division for the EC-IKI PNJ 2024 Public Lecture.

SMAN 64 Jakarta – East Jakarta

2019 – 2022

High School – Science Program (MIPA)

KEY PROJECTS

- DC Motor Control Trainer Kit (2025–2026): Web monitoring (JavaScript ES6, HTML5, Tailwind CSS v4) and interactive 3D Digital Twin (Three.js r160) synchronized in real time with ESP32 hardware via Firebase Realtime Database.
- IoT Web Monitoring Dashboard (2025): Monitors temperature, soil moisture, and humidity using a MySQL database.
- Plant Monitoring Robot (2025): ESP32-C3 with DHT11 and soil moisture sensors, IoT-based.
- Logic Gate Training Module (2024): Practice unit for AND, OR, NOT, NAND, NOR, XOR, Full Adder, Multiplexer, Demultiplexer, Flip-flop, and BCD-to-7-Segment Decoder.
- 9V Power Supply with IC 7809 (2024): Designed, assembled, and tested a stable power supply for small-scale electronic devices.

SKILLS

- IoT & Microcontrollers: ESP32, Arduino, Raspberry Pi, Firebase (Realtime Database & Hosting).
- Control & Automation Systems: PID, Fuzzy Logic Controller, PLC programming, SCADA (Schneider Electric, Vijeo Citect).
- Protection Systems & Testing: Protection Relays (OCR, Differential, REF MiCOM), OMICRON TestUniverse, GOOSE Configuration.
- Programming: Python, C/C++, HTML, CSS, JavaScript, TypeScript, React, Node.js, Express.
- Design & Simulation: NI Multisim, Proteus, AutoCAD, Fusion 360, Eagle.
- Databases: MySQL, PostgreSQL.
- Design & Multimedia: Adobe Photoshop, Adobe Lightroom, Canva, CapCut.
- Microsoft Office: Word, Excel, PowerPoint.
- Languages: Indonesian (native), Malay, English (limited but functional).